

WEEK 8:

MULTINOMIAL LOGIT REGRESSION

APPLIED STATISTICAL ANALYSIS II

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ROADMAP THROUGH STATS LAND

Where we've been:

- Over-arching goal: We're learning how to make inferences about a population from a sample
- Before reading week: We learned how to assess the model fit of logit regressions

Today we will learn how to:

- Estimate and interpret linear models when our outcome is categorical

LOGIT REGRESSION TO MULTINOMIAL REGRESSION

For a binary outcome we estimate a logit regression as...

$$\text{logit}[P(Y_i = 1|X_i)] = \ln \left(\frac{P(Y_i = 1|X_i)}{1 - P(Y_i = 1|X_i)} \right) = \beta_0 + \beta_1 X_i$$

- Intercept (β_0): When $x = 0$, the odds that $Y=1$ versus the baseline $Y = 0$ are expected to be $\exp(\beta_0)$
- Slope (β_1): When x increases by one unit, the odds that $Y=1$ versus the baseline $Y = 0$ are expected to multiply by a factor of $\exp(\beta_1)$

► Review of logistic regression estimation

How do we alter this framework for an outcome with more than two categories?

MULTINOMIAL RESPONSE VARIABLE

- Suppose we have a response variable that can take three possible outcomes that are coded as "1", "2", "3"
- More generally, the outcome y is categorical and can take values $1, 2, \dots, k$ such that ($k > 2$)

$$P(Y = 1) = p_1, P(Y = 2) = p_2, \dots, P(Y = k) = p_k$$

$$\sum_{j=1}^k p_j = 1$$

MULTINOMIAL REGRESSION: ESTIMATION

So, let "1" be the baseline category

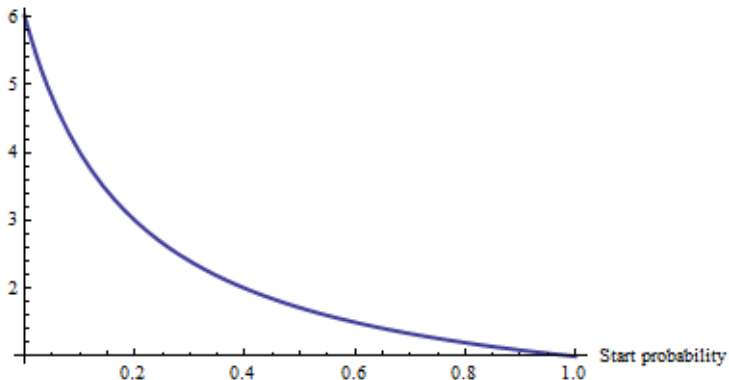
$$\ln \left(\frac{p_{i2}}{p_{i1}} \right) = \beta_{02} + \beta_{12}X_i$$

$$\ln \left(\frac{p_{i3}}{p_{i1}} \right) = \beta_{03} + \beta_{13}X_i$$

HOW TO THINK ABOUT MULTIPLYING ODDS

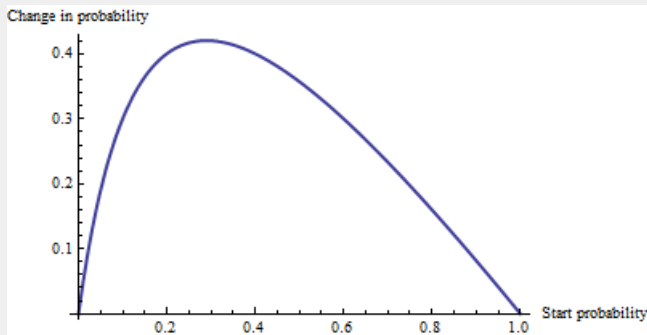
Starting odds	Ending odds	Starting $\Pr[Y=1]$	Ending $\Pr[Y=1]$
0.0001	0.0006	0.0001	0.0006
0.001	0.006	0.001	0.006
0.01	0.06	0.01	0.057
0.1	0.6	0.091	0.38
1.	6.	0.5	0.9
10.	60.	0.91	1.
100.	600.	0.99	1.

Probability ratio



HOW TO THINK ABOUT MULTIPLYING ODDS

- As an additive change, not much of a difference between a probability of 0.0001 and 0.0006 (it's only 0.05%)
- Also not much difference between 0.99 and 1 (only 1%)
- Largest additive effect occurs when the odds equal $\frac{1}{\sqrt{\beta}}$, where the probability changes from 29% to 71% ($\uparrow$ 42%)



EXAMPLE: EXPANSION OF RELIGIOUS GROUPS

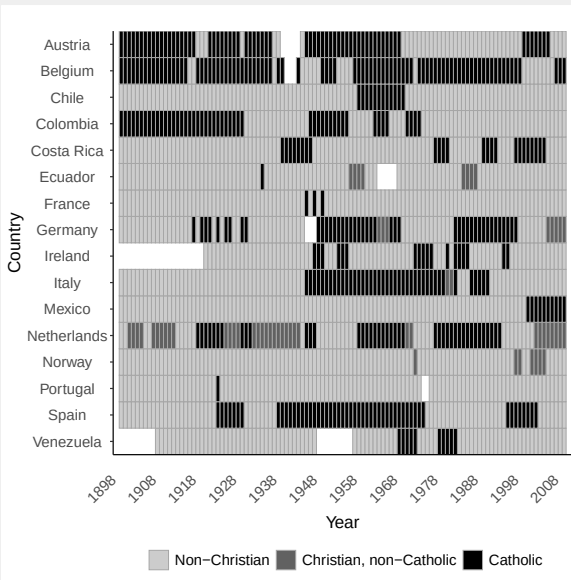
When and where do religious groups expand?

- Religious marketplace: # of followers, competition from other denominations
- Political marketplace: Government policies around religion

Case study: Expansion of dioceses in Catholic Church (1900-2010)

- Outcome: Diocese change in a given country (-, no Δ , +)
- Predictors:
 - ▶ % of population that is Catholic
 - ▶ % of pop. that is Christian, but not Catholic (Protestant, Evangelical, etc.)
 - ▶ Government religiosity: Non-Christian, Christian, Non-Catholic, Catholic

VARIATION IN GOVERNMENT RELIGIOUSITY



MULTINOMIAL REGRESSION: ESTIMATION IN R

```
1 # import data
2 diocese_data <- read.csv("https://raw.githubusercontent.com/ASDS-TC/StatsII_Spring2024/
  main/datasets/diocese_data.csv", stringsAsFactors = F)

1 # run base multinomial logit
2 multinom_model1 <- multinom(state3 ~ cathpct_lag + hogCatholic_lag +
3   propenpt_lag + country,
4   data = diocese_data)
```

	Decrease	Increase
% Catholic	-0.069** (0.025)	-0.007 (0.012)
% Protestant & Evangelical	-0.035 (0.042)	0.033* (0.015)
Government _{Catholic}	0.579 (0.504)	0.519* (0.234)
Government _{Christian,non-Catholic}	-148.089 (0.646)	-0.151 (0.862)
Constant	2.128 (2.226)	-0.839 (1.192)
Deviance	1896.567	1896.567
N	2808	2808

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Now what?

MULTINOMIAL REGRESSION: INTERPRETATION

- For every one unit increase in X , the **log-odds** of $Y = j$ vs. $Y = 1$ increase by β_{1j}
- For every one unit increase in X , the **odds** of $Y = j$ vs. $Y = 1$ multiply by a factor of $\exp(\beta_{1j})$

$$\ln \left(\frac{p_{i\text{Increase}}}{p_{i\text{None}}} \right) = -0.839 + 0.33X_{\%Protestant} + 0.519X_{\text{Gov't Catholic}} - 0.151X_{\text{Gov't Christian}} - 0.007X_{\%Catholic}$$

$$\ln \left(\frac{p_{i\text{Decrease}}}{p_{i\text{None}}} \right) = 2.128 - 0.35X_{\%Protestant} + 0.579X_{\text{Gov't Catholic}} - 148.089X_{\text{Gov't Christian}} - 0.069X_{\%Catholic}$$

MULTINOMIAL REGRESSION: INTERPRETATION

```
1 # exponentiate coefficients
2 exp(coef(multinom_model1)[,c(1:5)]))
```

	(Intercept)	% Catholic	Catholic Gov't	Christian, non-Catholic Gov't	% Protestant
Decrease	8.40	0.93	1.78	0.00	0.97
Increase	0.43	0.99	1.68	0.86	1.03

- So, in a given country, there is an increase in the baseline odds that the Church will expand by 1.68 times when a government supports policies that the Church supports

MULTINOMIAL REGRESSION: PREDICTION

For $j = 2, \dots, k$, we calculate the probability p_{ij} as

$$p_{ij} = \frac{\exp(\beta_{0j} + \beta_{1j}x_i)}{1 + \sum_{j=2}^k \exp(\beta_{0j} + \beta_{1j}x_i)}$$

For baseline category ($j = 1$) we calculate probability (p_{i1}) as

$$p_{i1} = 1 - \sum_{j=2}^k p_{ij}$$

We'll use these probabilities to assign a category of the response for each observation

PREDICTION IN R: CREATING DATA

```
1 # create hypothetical cases for predicted values
2 predict_data <- data.frame(hogCatholic_lag = rep(c("Non-Christian", "Catholic", "
   Christian, non-Catholic"), each = 3), cathpct_lag = rep(c(25, 50, 75), 3),
3                               propenpt_lag = rep(c(75, 50, 25), 3), country="Spain")
```

	hogCatholic_lag	cathpct_lag	propenpt_lag	country
1	Non-Christian	25.00	75.00	Spain
2	Non-Christian	50.00	50.00	Spain
3	Non-Christian	75.00	25.00	Spain
4	Catholic	25.00	75.00	Spain
5	Catholic	50.00	50.00	Spain
6	Catholic	75.00	25.00	Spain
7	Christian, non-Catholic	25.00	75.00	Spain
8	Christian, non-Catholic	50.00	50.00	Spain
9	Christian, non-Catholic	75.00	25.00	Spain

PREDICTION IN R: EXTRACT PREDICTED VALUES

```
1 # store the predicted probabilities for each covariate combo
2 predicted_values <- cbind(predict_data, predict(multinom_model1, newdata = predict_data,
  type = "probs", se = TRUE))
3
4 # calculate the mean probabilities within each level of government religiosity
5 by(predicted_values[, 5:7], predicted_values$hogCatholic_lag, colMeans)
```

predicted_values\$hogCatholic_lag: Catholic

None Decrease Increase

0.45543215 0.04051514 0.50405271

predicted_values\$hogCatholic_lag: Christian, non-Catholic

None Decrease Increase

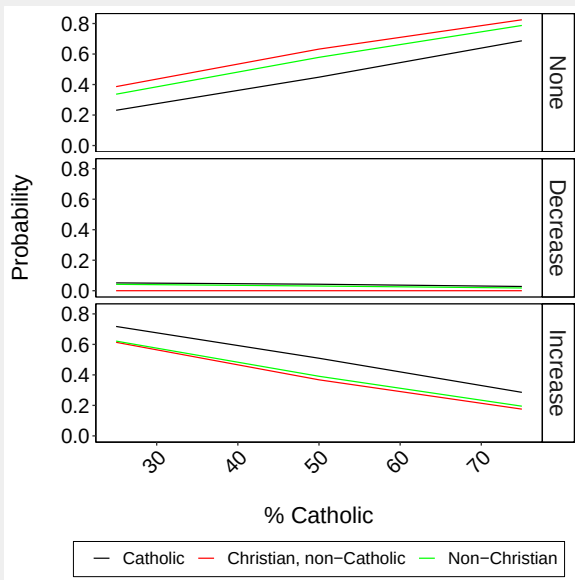
6.144487e-01 1.621855e-66 3.855513e-01

predicted_values\$hogCatholic_lag: Non-Christian

None Decrease Increase

0.56744365 0.03013509 0.40242126

PREDICTION IN R: PLOT



ACCURACY OF PREDICTIONS

```
1  
2 # see how well our predictions were:  
3 addmargins(table(diocese_data$state3, predict(multinom_  
    model1, type="class")))
```

	None	Decrease	Increase	Sum
None	2290.00	0.00	59.00	2349.00
Decrease	26.00	0.00	2.00	28.00
Increase	290.00	0.00	141.00	431.00
Sum	2606.00	0.00	202.00	2808.00

Thought for later: Not great prediction accuracy, maybe we want a different model?

POTENTIAL PROBLEMS: FOCUSING ON COEFFICIENTS

- Can lead researchers to overlook significant effects that affect their conclusions¹
 - ▶ Ex: Evaluate effect of information from events and messages on individuals' beliefs about war in Iraq (Gelpi 2017)²
 - Participants in experiment received different events and cues
 - "Events" treatment: subjects read a newspaper story with either "good news", "bad news", or a control condition with a news story that has no news events
 - "Opinions" treatment: newspaper story includes a confident statement from President Bush, cautious statements, or none
 - Outcome: Opinion on a scale regarding whether (1) surge was a success, (2) US would succeed in Iraq, and (3) should be a timetable for withdrawal

¹Paolino, Philip. "Predicted Probabilities and Inference with Multinomial Logit." *Political Analysis* 29.3 (2021): 416-421.

²Gelpi, Christopher. "The Surprising Robustness of Surprising Events: A Response to a Critique of "Performing on Cue"." *Journal of Conflict Resolution* 61.8 (2017): 1816-1834.

POTENTIAL PROBLEMS: FOCUSING ON COEFFICIENTS

- Original findings: "little evidence of any impact for elite opinion at all in this experiment"
 - ▶ Only coefficient that was "statistically significant": respondents who "strongly approved" of Bush + cautious message from Bush ↑ "somewhat disapprove" of timetable for withdrawal
- But, predicted probs show more nuanced story:

Table 1. Positive events, cautious cues, and a timetable for withdrawal in Iraq.

Timetable	Approval of President Bush			
	Strongly disapprove	Somewhat disapprove	Somewhat approve	Strongly approve
Positive events treatment				
Strongly approve	.060 (.056)	-.058 (.080)	-.001 (.028)	-.063 (.053)
Somewhat approve	-.117 (.057)	-.120 (.078)	-.013 (.051)	.103 (.100)
Somewhat disapprove	-.011 (.044)	.083 (.099)	-.089 (.053)	-.047 (.037)
Strongly disapprove	.068 (.034)	.095 (.106)	.104 (.070)	.007 (.113)
Cautious Bush treatment				
Strongly approve	-.034 (.052)	.008 (.088)	.006 (.024)	.025 (.089)
Somewhat approve	-.036 (.059)	.142 (.109)	.014 (.051)	-.036 (.037)
Somewhat disapprove	.081 (.054)	-.075 (.065)	.157 (.076)	.294 (.129)
Strongly disapprove	-.011 (.017)	-.076 (.089)	-.177 (.073)	-.283 (.130)
Observations	450	145	281	118

POTENTIAL PROBLEMS: INTERPRETATION TO BASELINE

- Sometimes you use specific baseline because RQ concerns change in occurrence of one outcome rather than another
 - ▶ If interested in relative odds, may be better off using coefficients for inference
 - ▶ But, should still calculate changes in predicted probabilities of relevant outcomes to understand changes in underlying probabilities
 - ▶ A significant log odds ratio resulting from a large change in probability of only one outcome may produce a substantively different interpretation than one where both probabilities change significantly

POTENTIAL PROBLEMS: INTERPRETATION TO BASELINE

Ex: Effect of trust upon acceptance of rumors in conflict zones
(Greenhill and Oppenheim 2017)³

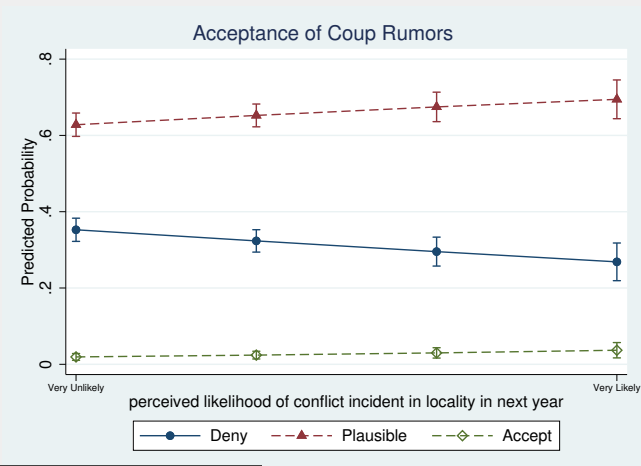
- Assess hypothesis that "as distrust of implicated entity rises, more likely it is that a rumor will be perceived as possibly or definitely true" against a baseline of disbelief (663)

³Greenhill, Kelly M., and Ben Oppenheim. "Rumor has it: The adoption of unverified information in conflict zones." *International Studies Quarterly* 61.3 (2017): 660-676.

POTENTIAL PROBLEMS: INTERPRETATION TO BASELINE

Interpretation of risk ratio, however, can be misleading

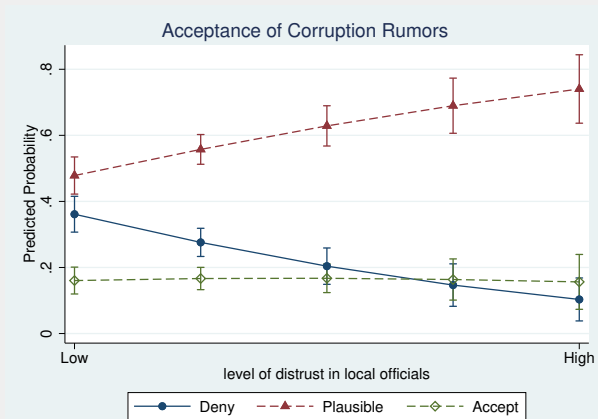
- In Thailand, little actual change in $\Pr(\text{plausible rumor})$ or $\Pr(\text{accept rumor})$



POTENTIAL PROBLEMS: INTERPRETATION TO BASELINE

In Philippines, both coefficients of distrust on beliefs about a rumor about corruption are statistically significant

- But, no change in accepting rumor is true given changes in respondents' trust in local officials



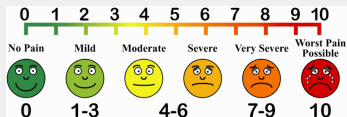
POTENTIAL PROBLEMS WITH MNL COEFFICIENTS

- Interpretations of odds against a baseline often imply significant coefficient = change in probabilities of both alternatives
 - ▶ But, change predicted probability of 1 alternative with a significant coefficient may be no different than change in predicted probability with a non significant coefficient
 - ▶ Because we often rely upon significance levels for testing hypotheses, demonstrates importance of examining changes in predicted probabilities
 - Even when you're interested only in effect of a covariate on odds of one category against a baseline

UNORDERED TO ORDERED MULTINOMIAL REGRESSION

- With ordinal data, let $Y_i = j$ if individual i chooses option j
- $j = 0, \dots, J$, so we have $J + 1$ choices
- Examples

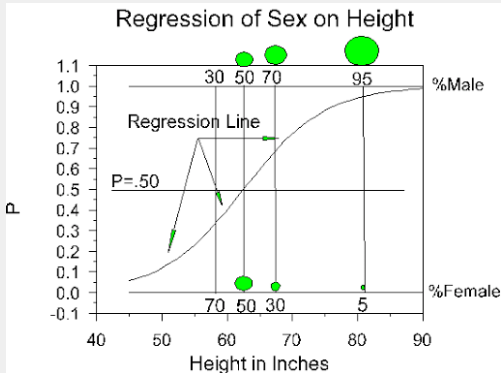
- ▶ Political economy: Budgets ($\downarrow$, no Δ , $\uparrow$)



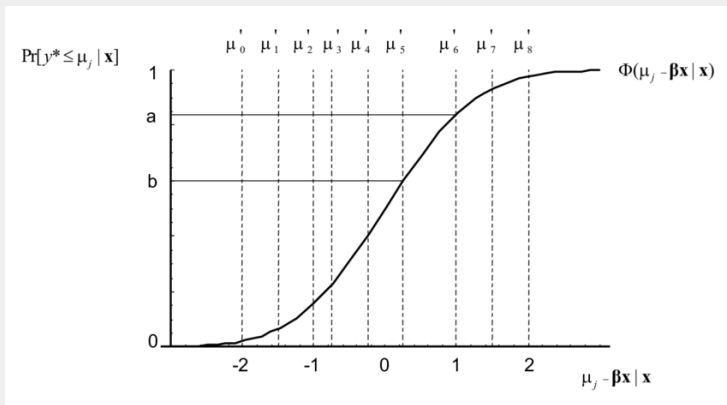
- ▶ Medicine: Patient pain
- ▶ Biology: Animal extinction (Extinct, Endangered, Protected, No regulation)
- How should we model $Pr(Y_i = j)$ to get a model we can estimate?
- Answer: $Pr(Y_i = j)$ will be areas under a probability density function (pdf)
 - ▶ Normal pdf gives ordered probit, logistic gives ordered logit

ESTIMATING CUTOFFS: INTUITION WITH LOGIT

- Binary dependent variable Y
= 0 or 1 (e.g., Male/Female)
- Estimate
 $\text{logit}[P(Y_i = 1|X_i)] =$
 $\ln\left(\frac{P(Y_i=1|X_i)}{1-P(Y_i=1|X_i)}\right) = \beta_0 + \beta_1 X_i$
- Calculate
 $X_i\beta = \hat{y} = \beta_0 + \beta_1 X_i + \dots$ for
each observation
- For observation i , if $\hat{y} < 0.5$
then predict $Y_i = 0$; if $\hat{y} > 0$
predict $Y_i = 1$



ESTIMATING CUTOFFS FOR ORDERED LOGIT



- For instance, if $b < X_i\beta < a$ then predict $Y_i = \mu_6 - \mu_5$
- Interpretation of β_1 : $\uparrow x_1$ by one unit changes the probability of going to next μ_j by β units
- The impact of this on $Pr(Y = 1)$ depends on your starting point

EXAMPLE: CONCERN ABOUT COVID-19 IN U.S.

How concerned are you that you or someone you know will be infected with the coronavirus?

- ABC News, web-based, nationally representative of U.S. (Beginning of pandemic; April 15 - 16, 2020)
- Political Hypothesis: Party ID predicts concern (Republicans less concerned than Democrats)

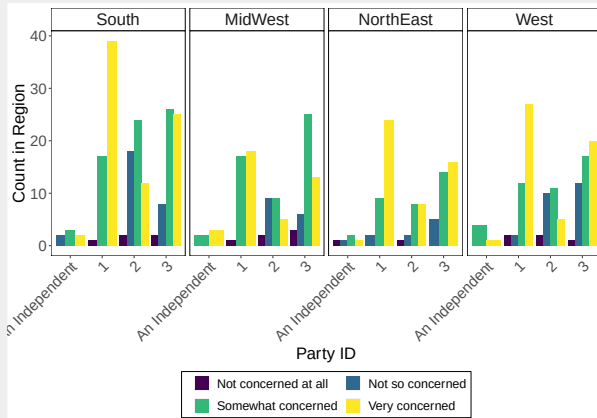
Potential confounders:

- Personal risk: Age
- Personal hygiene: "In the past week have you worn a face mask or face covering when you've left your home, or not" [Yes/No]
- Level of potential local infection: Population density [Metro/rural]
- Public health policies: Region [South, Northeast, Midwest, West]

COVID-19 CONCERN OVER TIME, BY PARTY, REGION...

2. How concerned are you that you or someone you know will be infected with the coronavirus?

	April 15-16	April 8-9	April 1-2	March 18-19	March 11-12
Very concerned	41	43	50	34	26
Somewhat concerned	40	43	39	45	40
Not so concerned	15	11	9	16	26
Not concerned at all	5	3	2	5	7
Skipped	-	-	1	*	1
Total concerned (Net)	80	86	89	79	66
Total not concerned (Net)	20	14	11	21	34



ORDERED MULTINOMIAL: ESTIMATION IN R

```
1 exp(cbind(OR = coef(ordered_logit), confint(ordered_logit)))
```

Did not leave home in the past week	1.052*** (0.306)
Wear mask outside	1.271*** (0.206)
Democrat	0.970*** (0.208)
Republican	-0.846*** (0.215)
Metro area	-0.274 (0.460)
MidWest	-1.369* (0.571)
NorthEast	0.110 (0.712)
West	-0.813 (0.758)
Metro area × MidWest	1.318* (0.627)
Metro area × NorthEast	0.077 (0.763)
Metro area × West	0.475 (0.793)
Deviance	1041.209
Num. obs.	514

***p < 0.001, **p < 0.01, *p < 0.05

What's missing? Intercept and cutoffs! Don't worry, we'll come back to this...

ORDERED MULTINOMIAL: INTERPRETATION

For now, how do we interpret the estimated coefficients? Similar procedure to multinomial logit!

- For every one unit increase in X , the **log-odds** of $Y = j$ vs. $Y = j + 1$ increase by β_1
- For every one unit increase in X , the **odds** of $Y = j$ vs. $Y = j + 1$ multiply by a factor of $\exp(\beta_1)$

$$\ln \left(\frac{p_{ij+1}}{p_{ij}} \right) = 1.052 \times \text{No leave} + 1.271 \times \text{Wear mask} + 0.970 \times \text{Democrat} + \dots + 0.475 \times \text{Metro area} \times \text{West}$$

ORDERED MULTINOMIAL: INTERPRETATION

```
1 # get odds ratios and CIs
2 exp(cbind(OR = coef(ordered_logit), confint(ordered_logit)))
```

	OR	2.5 %	97.5 %
Did not leave home in the past week	2.65	1.45	4.90
Wear mask outside	3.52	2.35	5.30
Democrat	2.65	1.76	4.02
Republican	0.43	0.28	0.66
Metro area	0.76	0.30	1.85
MidWest	0.25	0.08	0.77
NorthEast	1.12	0.28	4.59
West	0.44	0.10	1.99
Metro area×MidWest	3.74	1.10	12.98
Metro area×NorthEast	1.08	0.24	4.82
Metro area×West	1.61	0.33	7.67

- Sanity ✓: For those who take protective measures (staying at home, wearing a mask outside), odds of being more concerned about COVID are 2.6 to 3.5× higher compared to those who do not leave home or wear masks outside, holding constant all other variables

ORDERED MULTINOMIAL: FURTHER INTERPRETATION

	OR	2.5 %	97.5 %
Did not leave home in the past week	2.65	1.45	4.90
Wear mask outside	3.52	2.35	5.30
Democrat	2.65	1.76	4.02
Republican	0.43	0.28	0.66
Metro area	0.76	0.30	1.85
MidWest	0.25	0.08	0.77
NorthEast	1.12	0.28	4.59
West	0.44	0.10	1.99
Metro area $\times$ MidWest	3.74	1.10	12.98
Metro area $\times$ NorthEast	1.08	0.24	4.82
Metro area $\times$ West	1.61	0.33	7.67

- So, for a Democrat respondent in a given region, the odds of being more concerned about COVID are $2.65 \times$ higher compared to an Independent respondent, holding constant all other variables
- Geography: Population density and region

MODEL ASSUMPTIONS: PROPORTIONAL ODDS

Major assumption underlying ordinal logit (& ordinal probit) regression is that relationship between each pair of outcome groups is equal

- In other words, ordinal logistic regression assumes coefficients that describe relationship between lowest versus all higher categories of response variable are same as those that describe relationship between next lowest category and all higher categories, etc.
- Called proportional odds assumption or "parallel regression assumption"
- Because relationship between all pairs of groups is same, there is only one set of coefficients

CHECKING THE PARALLEL LINES ASSUMPTION

Estimate and compare simplified model:

```
1 # run logit regressions for all outcome categories
2 for(i in 1:length(unique(covid_data$covid_concern))){
3   assign(paste("logit_model", i, sep=""), glm(ifelse(covid_concern==unique(covid_data$covid_concern)[i], 1, 0) ~ mask + partyID, data = covid_data), envir=globalenv())
4 }
```

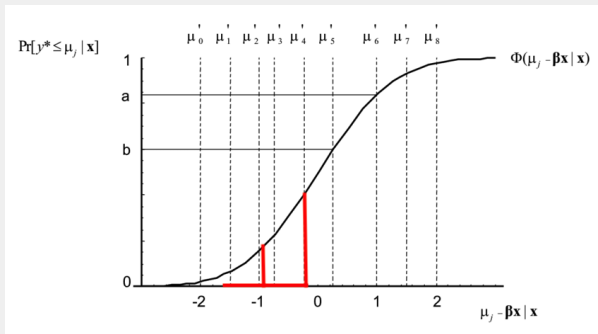
	Not concerned at all	Not so concerned	Somewhat concerned	Very concerned
Did not leave home in the past week	-0.014 (0.029)	-0.169** (0.053)	-0.007 (0.076)	0.190** (0.072)
Wear mask outside	-0.073*** (0.019)	-0.106** (0.036)	-0.085 (0.051)	0.264*** (0.048)
Democrat	-0.000 (0.019)	-0.121*** (0.035)	-0.101* (0.050)	0.222*** (0.047)
Republican	0.022 (0.020)	0.139*** (0.038)	-0.026 (0.054)	-0.134** (0.051)
(Intercept)	0.078*** (0.018)	0.242*** (0.034)	0.483*** (0.049)	0.197*** (0.046)
Deviance	16.729	58.186	120.155	106.944
Num. obs.	514	514	514	514

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Doesn't look like parallel odds holds, maybe not ordered categories (or wrong categories)?

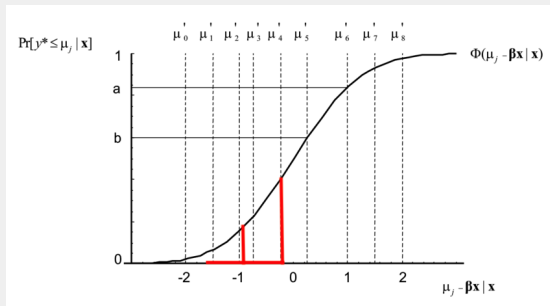
ORDERED MULTINOMIAL: PREDICTION

Remember those cutoffs?! This is where they're useful



- Assume Y has more than two ordered categories (for instance, low, medium, high)
- We now need two cut-points to divide the curve into three sections
- We'll estimate these as μ_1 and μ_2 using MLE

ORDERED MULTINOMIAL: PREDICTION



- If $X_i \beta < \mu_1$ then predict $Y_i = \text{low}$
- If $\mu_1 < X_i \beta < \mu_2$ then predict $Y_i = \text{medium}$
- If $X_i \beta > \mu_2$ then predict $Y_i = \text{high}$

$\beta \Delta$ in x may change the prediction on Y , or it may not!

PREDICTION IN R: CREATING DATA

```
1 # create fake data to use for prediction
2 predict_data <- data.frame(mask = rep(c("No", "Yes"), 3),
3   partyID = rep(c("An Independent", "A Democrat", "A Republican"), each = 2))
```

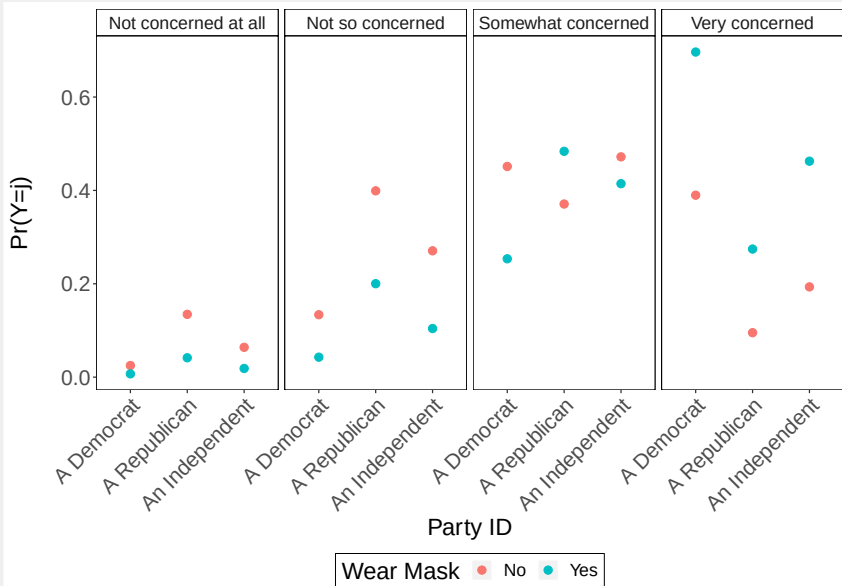
	Wear mask	party ID
1	No	An Independent
2	Yes	An Independent
3	No	A Democrat
4	Yes	A Democrat
5	No	A Republican
6	Yes	A Republican

PREDICTION IN R: EXTRACT PREDICTED VALUES

```
1 # add in predictions for each observation in fake data
2 plot_data <- melt(cbind(predict_data, predict(ordered_logit_slim, predict_data, type = "
  probs")), id.vars = c("mask", "partyID"),
3   variable.name = "Level", value.name="Probability")
```

	Wear mask	party ID	Level of concern	$\Pr(Y = j)$
1	No	An Independent	Not concerned at all	0.06
2	Yes	An Independent	Not concerned at all	0.02
3	No	A Democrat	Not concerned at all	0.03
4	Yes	A Democrat	Not concerned at all	0.01
	:	:	:	:
24	Yes	A Republican	Very concerned	0.27

PREDICTION IN R: PLOT



ACCURACY OF PREDICTIONS

```
1 # see how well our predictions were:  
2 addmargins(table(covid_data$covid_concern, predict(ordered_  
  logit_slim, type="class")))
```

	Not concerned at all	Not so concerned	Somewhat concerned	Very concerned	Sum
Not concerned at all	0	6	7	5	18
Not so concerned	0	18	38	21	77
Somewhat concerned	0	11	102	87	200
Very concerned	0	7	54	158	219
Sum	0	42	201	271	514

ANOTHER EX: SOCIAL TRUST AND POLITICAL INTEREST

Is paying attention to politics associated with less trust in others?

- H_1 : $\uparrow$ people pay attention to negative political messages $\downarrow$ they trust others

```
1 # load in example 2 data:  
2 # ESS political and social trust  
3 # clean up data  
4 # need to have registered email  
5 set_email("me@tcd.ie")  
6 # import all rounds from IE  
7 ESS_data <- import_all_cnrrounds(country = "Ireland")
```

SOCIAL TRUST AND POLITICAL INTEREST: OUR MODEL

Outcome:

- `ppltrst`: Most people can be trusted or you can't be too careful
 - ▶ Values: Ordinal 0-11

Predictors:

- `polintr`: How interested in politics
 - ▶ Values: 1-4 (not at all, not really, somewhat, quite)
- `essround`: ESS round (time)

ESTIMATE MODEL OF TRUST AND POLITICAL INTEREST

```
1 # run ordered logit
2 ordered_trust <- polr(ppltrst ~ polintr + essround, data =
  merged_ESS, Hess=T)
3
4 # get odds ratios and CIs
5 exp(cbind(OR = coef(ordered_trust), confint(ordered_trust))
  )
```

	OR	2.5 %	97.5 %
polintr2	0.95	0.88	1.04
polintr3	0.76	0.69	0.83
polintr4	0.58	0.53	0.64
essround2	1.35	1.21	1.50
essround3	0.89	0.80	0.99
essround4	0.91	0.81	1.014
essround5	0.78	0.70	0.86
essround6	0.75	0.68	0.83
essround7	0.80	0.72	0.89
essround8	2.29	2.07	2.54
essround9	2.05	1.84	2.28

PREDICTED PROBABILITIES WITH CIs

Get predicted probs for each category for person with high political interest (4) from last wave

	$\hat{y}$	2.5%	97.5%
0	0.021	0.018	0.023
1	0.027	0.025	0.030
2	0.040	0.037	0.044
3	0.064	0.059	0.069
4	0.077	0.071	0.082
5	0.151	0.144	0.158
6	0.144	0.139	0.149
7	0.176	0.169	0.182
8	0.180	0.169	0.191
9	0.076	0.070	0.083
10	0.035	0.032	0.039
11	0.009	0.007	0.011

CHANGE IN PREDICTED PROBABILITIES WITH CIs

See how predicted probs change between two cases: low interest (1) to high (4) for wave 1

	Δ	2.5%	97.5%
0	-0.013	-0.015	-0.010
1	-0.016	-0.018	-0.013
2	-0.021	-0.024	-0.017
3	-0.028	-0.032	-0.023
4	-0.025	-0.029	-0.021
5	-0.028	-0.033	-0.022
6	-0.002	-0.005	0.001
7	0.026	0.021	0.030
8	0.055	0.046	0.064
9	0.031	0.025	0.037
10	0.016	0.013	0.019
11	0.004	0.003	0.005

WRAP UP

In this lesson, we went over how to...

- Extended logit regression for outcomes that have more than one category
- Extended multinomial logit regression for outcomes are ordered categories

In the coming weeks, we'll talk about event & count outcome variables

REVIEW: LOGIT REGRESSION ESTIMATION

We want to think about how to take our set up from a binary logit and apply it to multiple outcome categories

- Without any loss of generality, index the X_i so that X_k is the variable and we're still interested in taking exp of β_k
- Fixing the values of each covariate and varying X_k by a small amount yields some δ

$$\log(p(\dots, X_k + \delta)) - \log(p(\dots, X_k)) = \beta_k \delta$$

- Thus, β_k is marginal change in log odds with respect to X_k

REVIEW: LOGIT REGRESSION ESTIMATION

To recover $\exp(\beta_k)$, we must set $\delta=1$ and exponentiate left hand side:

$$\exp(\beta_k) = \exp(\beta_k) \times \delta \quad (1)$$

$$= \exp(\beta_k) \quad (2)$$

$$= \exp(\log(p(\dots, X_k + \delta)) - \log(p(\dots, X_k))) \quad (3)$$

$$= \frac{p(\dots, X_k + 1)}{p(\dots, X_k)} \quad (4)$$

$$(5)$$

► [Back to slides](#)